Figure 19.1. Integrated Concept Map: Artificial Intelligence, Machine Learning, and Emerging Neurotechnologies

Chapter topics and a reading framework

- 19.1 AI foundations
- 19.2 Supervised learning
- 19.3 Unsupervised learning
- 19.4 Deep learning
- 19.5 Neuroimaging classification
- 19.6 Digital phenotyping
- 19.7 Wearable sensing
- 19.8 Brain-computer interfaces
- 19.9 Explainability and bias
- 19.10 Responsible innovation

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 19, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.